
ET 501: Power Electronic Applications in Electric Transportation

Course Code	ET 501
Course Name	Power Electronic Applications in Electric Transportation
L-T-P-C	3-0-0-3
Prerequisites	Power Electronics/Consent of the instructor
Intended for	PG
Mutual Exclusion	None
Approval	44th BoA (Nov 2021)

Course Contents

- **Introduction to hybrid and electric vehicles** (3 hours)
 - Electrification concepts
 - HEV architectures and classifications
 - Technological trends
 - Electric drivetrains
- **Introduction to Power Electronics** (7 hours)
 - Basic power electronics concepts
 - Overview of power semiconductor devices
 - Various converters for EVs
- **Power electronics Converters** (12 hours)
 - DC-DC converters
 - AC-DC converters
 - DC-AC converters
- **Battery Connected Systems** (6 hours)
 - Battery pack
 - Battery management system
 - Thermal management system
 - Body control unit
- **Charging Infrastructure for EVs** (6 hours)
 - On-board charging
 - Fast charging
 - Battery-swapping station
 - DC-microgrid based charging station
- **Modelling and Simulation** (8 hours)
 - System design considerations
 - Rating and sizing of electric drivetrain components
 - Complete system modelling
 - Simulation of the complete system

Laboratory Modules

A laboratory course is proposed separately to support this course.

Text Books

1. A. Emadi, M. Ehsani and J. M. Miller, **Vehicular Electric Power Systems: Land Sea Air and Space Vehicles**, New York:Marcel Dekker, 2003.
2. J. Larminie and J. Lowry, **Electric Vehicle Technology Explained**, New York:Wiley, 2003.

References

1. Ned Mohan, Tore M. Undeland, William P. Robbins, **Power Electronics, Applications and Design**, John Wiley & Sons, 2003.
2. M.H. Rashid, **Power Electronics: Circuits, Devices and Application**, Fourth edition, Pearson Education, 2017
3. Sheldon S. Williamson, **Energy Management Strategies for Electric and Plug-in Hybrid Electric Vehicles**, 1st Edition, Springer, 2003.
4. B. K. Bose, **Modern Power Electronics and AC Drives**, Pearson Education India, 2015